

$$C_{ij} = (-1)^{i+j} M_{ij}$$

$$\det(A) = a_{1j} C_{1j} + \dots$$

1. จงแสดงวิธีทำและหาดีเทอร์มิแนนต์ของ

$$A = \begin{bmatrix} 2 & 0 & 1 & 3 & -2 \\ -2 & 1 & 3 & 2 & -1 \\ 1 & 0 & -1 & 2 & 3 \\ 3 & -1 & 2 & 4 & -3 \\ 1 & 1 & 3 & 2 & 0 \end{bmatrix} \quad \begin{aligned} & 0(C_{12}) + 1(C_{22}) + 0(C_{32}) + (-1)(C_{42}) + 1(C_{52}) \\ & = 0 + 13 + 0 - 115 - 33 \\ & = -135 \neq \end{aligned}$$

$$C_{22} = (-1)^4 \begin{vmatrix} 2 & 1 & 3 & -2 \\ 1 & -1 & 2 & 3 \\ 3 & 2 & 4 & -3 \\ 1 & 3 & 2 & 0 \end{vmatrix} \quad \begin{aligned} & = 1(-7) + 3(4) + 2(4) + 0 \\ & = -7 + 12 + 8 = 13 \end{aligned}$$

$$C_{41} = (-1)^5 \begin{vmatrix} 1 & 3 & -2 & 1 & 3 \\ -1 & 2 & 3 & -1 & 2 \\ 2 & 4 & -3 & 2 & 4 \end{vmatrix} \quad \begin{aligned} & = -6 + 18 + 8 + 8 + -12 + -9 \\ & = -7 \end{aligned}$$

$$C_{42} = \begin{vmatrix} 2 & 3 & -2 & 2 & 3 \\ 1 & 2 & 3 & 1 & 2 \\ 3 & 4 & -3 & 3 & 4 \end{vmatrix} \quad \begin{aligned} & = -12 + 27 - 8 + 12 - 24 + 9 \\ & = 4 \end{aligned}$$

$$C_{43} = -1 \begin{vmatrix} 2 & 1 & -2 & 2 & 1 \\ 1 & -1 & 3 & 1 & -1 \\ 3 & 2 & -3 & 3 & 2 \end{vmatrix} \quad \begin{aligned} & = 6 + 9 - 4 - 6 - 12 + 3 \\ & = -1(-4) = 4 \end{aligned}$$

$$C_{42} = \begin{vmatrix} 2 & 1 & 3 & -2 \\ -2 & 3 & 2 & -1 \\ 1 & -1 & 2 & 3 \\ 1 & 3 & 2 & 0 \end{vmatrix} \quad \begin{aligned} & = 1(32) + 3(43) + 2(-23) \\ & = -22 + 129 + -46 = 115 \end{aligned}$$

$$C_{41} = -1 \begin{vmatrix} 1 & 3 & -2 & 1 & 3 \\ 3 & 2 & -1 & 3 & 2 \\ -1 & 2 & 3 & -1 & 2 \end{vmatrix} \quad \begin{aligned} & = 6 + 3 - 12 - 4 + 2 - 27 \\ & = (-1)(-32) = 32 \end{aligned}$$

$$C_{42} = \begin{vmatrix} 2 & 3 & -2 & 2 & 3 \\ -2 & 2 & -1 & -2 & 2 \\ 1 & 2 & 3 & 1 & 2 \end{vmatrix} \quad \begin{aligned} & = 12 - 3 + 8 + 4 + 4 + 18 \\ & = 43 \end{aligned}$$

$$C_{43} = -1 \begin{vmatrix} 2 & 1 & -2 \\ -2 & 3 & -1 \\ 1 & -1 & 3 \end{vmatrix} \begin{vmatrix} 2 & 1 \\ -2 & 3 \\ 1 & -1 \end{vmatrix} = 18 - 1 - 4 + 6 - 2 + 6 = -23$$

6 -2 6
18 -1 -4

$$C_{52} = -1 \begin{vmatrix} 2 & 1 & 3 & -2 \\ -2 & 3 & 2 & -1 \\ 1 & -1 & 2 & 3 \\ 3 & 2 & 4 & -3 \end{vmatrix} = -62 + 12 - 3 + 86 = -1(33) = -33$$

$$C_{12} \begin{vmatrix} -2 & 2 & -1 \\ 1 & 2 & 3 \\ 3 & 4 & -3 \end{vmatrix} \begin{vmatrix} -2 & 2 \\ 1 & 2 \\ 3 & 4 \end{vmatrix} = 12 + 18 - 4 + 6 + 24 + 6 = -1(62) = -62$$

6 24 6
12 18 -4

$$C_{22} \begin{vmatrix} 2 & 3 & -2 \\ 1 & 2 & 3 \\ 3 & 4 & -3 \end{vmatrix} \begin{vmatrix} 2 & 3 \\ 1 & 2 \\ 3 & 4 \end{vmatrix} = -12 + 27 - 8 + 12 - 24 + 9 = 4$$

12 -24 9
-12 27 -8

$$C_{32} \begin{vmatrix} 2 & 3 & -2 \\ -2 & 2 & -1 \\ 3 & 4 & -3 \end{vmatrix} \begin{vmatrix} 2 & 3 \\ -2 & 2 \\ 3 & 4 \end{vmatrix} = -12 - 9 + 16 + 12 + 8 - 18 = -1(-3) = 3$$

12 8 -18
-12 -9 16

$$C_{42} \begin{vmatrix} 2 & 3 & -2 \\ -2 & 2 & -1 \\ 1 & 2 & 3 \end{vmatrix} \begin{vmatrix} 2 & 3 \\ -2 & 2 \\ 1 & 2 \end{vmatrix} = 12 - 3 + 8 + 4 + 4 + 18 = 43$$

4 4 18
12 -3 8

$$A^{-1} = \frac{1}{\det(A)} \times \text{adj.}$$

2. จงแสดงวิธีทำและหาค่า $|A^{-1}|$

นั่นคือ เริ่มต้นด้วยการหา A^{-1} จากนั้นให้ทำการหาดีเทอร์มิแนนต์ของเมทริกซ์นั้น

$$A = \begin{bmatrix} 1 & 0 & -1 & 3 \\ 1 & 0 & 3 & -2 \\ 2 & 0 & 2 & -1 \\ 1 & -3 & 1 & 2 \end{bmatrix}$$